Lab visit checklist: chip flow angle dataset

You get one visit, so this is ordered by priority. If you run short of time, the top of each section matters most.

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1. Ask the lab staff first (5 min, high value)

- □ **The CNC programs (G-code) for the videos already shot on 12–13 Oct.** Photograph the program on the controller screen. It contains spindle speed (S), feed (F), and depths per pass. With it, the videos you already have can get parameters attached.
- □ **Workpiece material grade**, e.g. Al 6061-T6, 6063, 2024. "Aluminium" is not enough.
- □ **Insert ISO code** (e.g. `CCMT 09T304-xx`) and **chip-breaker name**. Photograph the insert box or label.
- □ **Tool holder ISO code** (e.g. `SCLCR 2020K09`). Photograph the marking on the holder. It gives the entering/approach angle, rake and inclination angle.
- □ Machine make/model and max RPM.
- □ Whether the controller shows **spindle load %**. If yes, film the screen during each run too (free extra feature).
- □ Permission and the safe place to mount a phone.

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2. Parameters to record for EVERY run

Field	Notes
Run ID	R01, R02, … (also shown on a paper slate in the video)
Video filename(s)	e.g. IMG_1410.MOV. Note it right after recording
Operation	OD turning / facing / grooving / profiling
Feed direction	towards chuck / away / radial in / radial out
Speed	Write **N (rpm)** AND whether the program uses **G96 constant surface speed (Vc, m/min)** or **G97 fixed rpm**
Diameter at cut	start and end (mm). Needed to compute Vc = π·D·N/1000
Feed f	mm/rev
Depth of cut ap	mm (radial for OD turning, axial for facing)
Insert + nose radius rε	from the ISO code (last digits, e.g. 04 = 0.4 mm)
Chip breaker	name/code
Holder	ISO code (entering angle κr)
Cutting edge	new / used (which corner)
Coolant	off / flood / mist, and type
Chip form seen	continuous, helical, C-shaped, broken, **tangled**
Entangled?	yes / no, and around what (workpiece, tool, chuck)
Notes	anything odd

The 3 most important are feed, depth of cut and speed, but chip *flow angle* also depends strongly on **nose radius, entering angle and inclination angle**. Without the tool geometry, the model cannot generalise.

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3. Experiment plan (what to vary)

The theory says depth of cut and feed (relative to the nose radius) drive the flow angle most, and speed less. So:

Minimum plan (~13 runs, same tool):

- 3 feeds × 3 depths of cut at one speed = 9 runs (e.g. f = 0.08 / 0.15 / 0.25 mm/rev, ap = 0.3 / 1.0 / 2.0 mm, chosen with the operator so they're safe for the machine)
- 2 extra speeds at the middle feed/depth = 2 runs
- Repeat the middle point 2 more times = 2 runs (shows how noisy the measurement is)
- Do the runs in **shuffled order**, not sorted.

If there's more time:

- Repeat the 3×3 at a second speed.
- Change the insert to a different **nose radius** or holder (different entering angle) and repeat a few points. This is the most valuable addition.
- Try with and without coolant at the middle point.
- Try a depth of cut *smaller* than the nose radius. The flow angle changes most here.

Keep each run long enough: at least 10–15 s of steady cutting in the video.

3a. Cut length, recording time and depth of cut

Cut length: 10 cm per run (prof's recommendation). That's a good choice: it gives a long stretch of steady cutting, and long continuous chips have time to show whether they tangle.

Cutting time for 10 cm (OD turning):

> time (s) = 60 × 100 mm ÷ (feed f [mm/rev] × spindle speed N [rpm]) = **6000 ÷ (f × N)**

Spindle speed N	f = 0.08 mm/rev	f = 0.15 mm/rev	f = 0.25 mm/rev
800 rpm	94 s	50 s	30 s
1200 rpm	63 s	33 s	20 s
1600 rpm	47 s	25 s	15 s
2000 rpm	38 s	20 s	12 s

(If the operator sets cutting speed instead: N = 1000 × Vc ÷ (π × D). For example, Vc = 250 m/min on a 50 mm bar gives about 1600 rpm.)

Recording time per run = cutting time + about 5 s before the tool touches + about 5 s after it leaves. In practice that's **about 20 s to 100 s per run**. Start recording before the cycle start button is pressed,

and stop after the tool retracts.

Depth of cut: what I suggest

- Use 3 levels, set **relative to the insert's nose radius rε** (read it from the insert code):
  - **Low: just below rε**, e.g. **0.3 mm** for a 0.4 mm nose radius. The chip flow angle changes most in this range, so it's the most informative level.
  - **Medium: about 1.0 mm.**
  - **High: about 2.0 mm.**
- Check the insert box for its recommended ap and feed range (the chip breaker's working window), and stay inside it.
- The operator has the final say on what's safe for the machine and workpiece.

Stock budget (check this with the operator before starting): each 10 cm pass removes 2 × ap from the diameter. The minimum plan (13 runs) removes about **13–14 mm of radius, i.e. ~27 mm off the diameter.** A 50 mm bar would end up near 23 mm, which is thin for a long overhang. So either:

- use a bar of **≥ 60 mm** diameter with the tailstock centre engaged, or
- use **two bars**, or
- cut different 10 cm zones of a long bar.

Speed tip: because the diameter keeps shrinking, ask for **G96 (constant surface speed)** so the cutting speed Vc stays the same across runs. Still write down the diameter for every run.

Total time estimate: 13 runs × (~1 min of cutting + ~3–5 min for setting up, measuring and logging) ≈ **1–1.5 hours**, plus the camera setup and calibration photo.

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4. How to film (this decides whether the angle can be measured at all)

The current videos are handheld, with the camera angle, zoom and orientation changing. A 2D angle in the image then depends on where the phone was, so those angles **can't be compared across clips**. For the new data:

- □ **Phone fixed on a tripod/clamp/magnetic mount. Same position for ALL runs.** Don't touch it between runs. If it must move, start a new "camera setup" in the log.
- □ **View:** looking down onto the **rake face of the insert** (top of the insert), so the chip leaving the cutting edge is seen from above. If you have a **second phone**, add a side view along the workpiece axis.
- □ **Landscape**, no zoom changes, **slow-motion 120 or 240 fps** if the phone supports it (otherwise 60 fps, 1080p or 4K).
- □ **Lock focus and exposure** (long-press on the iPhone screen → AE/AF LOCK) on the tool tip.
- □ Turn off Cinematic / Action mode and auto macro switching.
- □ **Dark background** behind the chip if possible (black card/cloth out of the chip path). Shiny aluminium on shiny metal is hard to segment.
- □ Extra light from the side if it's dim, to avoid strong glare on the insert.
- □ **Calibration shot (machine STOPPED, with permission):** with the camera in its final position, photograph a **ruler** and a **protractor or printed angle grid** laid at the insert's height, plus the workpiece axis visible. This lets the image angle be converted to a real angle.
- □ **Slate:** before each run, hold a paper with the run ID in view for 1 s (or say it aloud).

- □ Film from **before the tool engages until after it exits**.
- □ Keep the phone safe from chips and coolant: behind the door glass or with a clear bag over it. Never reach in while the spindle is running.

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5. Physical chip samples (cheap, very useful)

- □ After each run, collect a few chips into a **labeled zip bag** (run ID).
- □ Photograph them on a dark sheet with a ruler.
- □ Later: measure **chip thickness with calipers/micrometer**. Chip thickness ÷ uncut thickness = chip compression ratio, another feature and a check on the physics.

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6. What to bring

- □ Tripod, clamp, or magnetic phone mount (+ a second phone if possible)
- □ Power bank and **at least 15 GB free storage**. iPhone Settings → Camera → Record Slo-mo shows the size per minute for each mode. The plan is ~15–20 min of footage.
- □ Black card/cloth, small torch or LED light
- □ Ruler, protractor (or printed angle grid), marker, tape
- □ Zip bags for chips
- □ This checklist + a printed copy of the log table in section 2
- □ Safety glasses, closed shoes; follow the lab's rules

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7. Before you leave the lab

- □ Every video filename is matched to a run ID in the log.
- □ You have photos of the insert/holder labels and the G-code (old and new runs).
- □ The calibration photo is taken from the final camera position.
- □ AirDrop/upload the videos right away (WhatsApp compresses video, so don't send through it).